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$$\lambda = h / mc \text{ ----------(2.8)}$$

The equation 2.8 represents the wavelength of photons whose momentum is given by mc (Photons have zero rest mass)

For a particle of matter with mass m and moving with a velocity v, the equation 2.8 can be written as

$$\lambda = h / mv \text{ ---------- (2.9)}$$

This is valid only when the particle travels at speeds much less than the speed of Light.

This equation implies that a moving particle can be considered as a wave and a wave can exhibit the properties (i.e momentum) of a particle. For a particle with high linear momentum (mv) the wavelength will be so small and cannot be observed. For a microscopic particle such as an electron, the mass is of the order of 10^{-31} kg, hence the wavelength is much larger than the size of atom and it becomes significant.

Let us understand this by calculating de Broglie wavelength in the following two cases:

(i) A 6.626 kg iron ball moving with 10 ms^{-1}

(ii) An electron moving at 72.73 ms^{-1}

$\lambda_{iron\ ball} = h/mv$

$$= \frac{6.626 \times 10^{-34} kgm^2s^{-1}}{6.626\ kg \times 10\ ms^{-1}} = 1 \times 10^{-35}\ m$$

$\lambda_{electron} = h/mv$

$$= \frac{6.626 \times 10^{-34} kgm^2s^{-1}}{9.11 \times 10^{-31}\ kg \times 72.73\ ms^{-1}}$$

$$= \frac{6.626}{662.6} \times 10^{-3}\ m = 1 \times 10^{5}\ m$$

For the electron, the de Broglie wavelength is significant and measurable while for the iron ball it is too small to measure, hence it becomes insignificant.

Evaluate Yourself

1. Calculate the de-Broglie wavelength of an electron that has been accelerated from rest through a potential difference of 1 keV.

2.2.1 Quantisation of angular momentum and de Broglie concept:

According to the de Broglie concept, the electron that revolves around the nucleus exhibits both particle and wave character. In order for the electron wave to exist in phase, the circumference of the orbit should be an integral multiple of the wavelength of the electron wave. Otherwise, the electron wave is out of phase.

Circumference of the orbit $= n\lambda$

$2\pi r = n\lambda$ ----------(2.10)

$2\pi r = nh/mv$

Rearranging, $mvr = nh/2\pi$ ----------(2.1)

Angular momentum $= nh/2\pi$

The above equation was already predicted by Bohr. Hence, De Broglie and Bohr's concepts are in agreement with each other.